Agent-GhibliLock: Deterministic Wavelet-Locked Cross-Modal Style Transfer, Sliced-Wasserstein Chromaticity Transport, and ISO/IEC 19566-5 C2PA JUMBF Provenance Compilation

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Abstract—Enterprise media, retail, and creative automation workflows increasingly deploy cross-modal Vision-Language-to-Diffusion pipelines ("Pixels $\rightarrow$ Prose $\rightarrow$ Pixels/Video"), wherein a Vision-Language Model (VLM) decomposes a source photograph into a structured visual inventory and style prompt (e.g., a Ghiblification Prompt, `GhP`) that conditions a generative image or video synthesis model paired with coarse spatial reference/pose guidance. Even when macro-level spatial registration is maintained upstream, routing visual state through natural language and stochastic latent decoding introduces an information-theoretic fine-phase bottleneck—*The Captioning Conundrum*—alongside cryptographic provenance severance: (1) high-frequency spatial phase geometry (such as corporate apparel emblems, diagonal foreground railings, and perspective vanishing lines) is eroded during latent decoding ($\text{SSIM}_{\text{edge}} = 0.0413$ under prose-only synthesis, and 0.4482 under coarse SDEdit); (2) multi-frame video generation across a Group of Pictures (GOP) exhibits inter-frame global luminance jump variance ($\sigma^2_{\text{flicker}} = 345.240$) and local subband pulsing ($\sigma^2_{\text{local}} = 33.349$); and (3) transitioning through text severs the Coalition for Content Provenance and Authenticity (C2PA) ingredient chain, while social/web gateways frequently strip outer container metadata chunks.

This paper presents **Agent-GhibliLock**, a deterministic post-synthesis signal-processing and cryptographic binary container compiler that integrates classical multiresolution and optimal-transport primitives to enforce structural phase fidelity, perceptual chromaticity alignment, and dual-layer provenance given upstream macro-spatial registration (within the 2^K-pixel Haar filter support). Prior to export, Agent-GhibliLock transforms the spatially registered source asset and diffusion candidate into perceptual CIELAB ($L^*a^*b^*$) space. First, an Iterative 3D Sliced-Wasserstein Optimal Transport (SWOT) solver aligns the candidate's 3D (L^*, a^*, b^*) distribution with a target artistic reference palette Y_{ref} over orthonormal $O(3)$ projections using rank-order permutation bijections (achieving pre-Step-B $\widehat{W}_2^{\text{StepA}} = 1.7638 \pm 0.0301$). Second, an orthonormal multi-level 2D Discrete Haar Wavelet Transform ($\text{DWT}_2^{(K)}$, $K = 2$) locks the candidate's high-frequency directional luminance detail subbands (LH_k, HL_k, HH_k) to the source photograph's phase geometry while retaining the SWOT-aligned coarse approximation subband (LL_K) and chrominance channels (with post-Step-B $\widehat{W}_2^{\text{StepB}} = 4.2886 \pm 0.3733$, post-QIM exported $\widehat{W}_2^{\text{out}} = 4.3071 \pm 0.3981$, and out-of-sample held-out $\widehat{SW}_2^{\text{eval}} = 4.6240 \pm 0.5084$ over $M_{\text{eval}} = 64$ independent directions). For video GOPs, reference-palette mean anchoring eliminates pre-clamping global frame-mean jump flicker ($\sigma^2_{\text{flicker}} = 0.0000$; 0.0001 post-sRGB rounding) while a wavelet-domain exponential moving average (EMA) filter ($\beta = 0.78$) attenuates local LL_K tile variance by 88.7% empirically (87.6% analytical steady-state bound). Finally, Agent-GhibliLock embeds a 64-bit provenance soft-binding digest into mid-band 8×8 Type-II 2D-DCT coefficients via Quantization Index Modulation (QIM) and compiles an ISO/IEC 19566-5 JUMBF (`caBX`) C2PA superbox into the ISO/IEC 15948 PNG container. Across $N = 600$ stratified 64×64 trials and live Google Cloud Vertex AI multi-model evaluations (128×128, $N_{\text{scenes}} = 10$ per tier), Agent-GhibliLock elevates $\text{SSIM}_{\text{edge}}$ to 0.9987 ± 0.0010 (100.00% pass rate) alongside non-circular full-band $\text{SSIM}_{\text{full}} = 0.9851$ and signed Sobel gradient correlation $\rho_{\text{grad}} = 0.8535$, and achieves 100.00% tamper localization and blind DCT watermark recovery (BER $= 0.0000$, PSNR > 47 dB vs. 24.16% BER for blind spread-spectrum DCT).

Index Terms—Cross-Modal Style Transfer, Discrete Wavelet Transform, Sliced-Wasserstein Optimal Transport, C2PA Provenance, ISO/IEC 19566-5 JUMBF, Quantization Index Modulation, Discrete Cosine Transform, Vision-Language Models, Diffusion Synthesis.

1 Introduction

The convergence of frontier Vision-Language Models (VLMs) and latent diffusion / autoregressive media synthesizers [1], [2] has catalyzed a new architectural paradigm in enterprise creative workflows: *Cross-Modal Visual-to-Text-to-Visual Compilation* ("Pixels $\rightarrow$ Prose $\rightarrow$ Pixels/Video"). While modern cloud synthesis APIs (such as Google Cloud Vertex AI `imagen-3.0-generate-002` and `veo-3.1-generate-001` within the Gemini multimodal ecosystem [1], [2]) expose native image-to-image (SDEdit-style [3]), spatial edge/pose (ControlNet [4]), and style-reference conditioning alongside text prompts, relying purely on high-strength pixel conditioning restricts bold artistic restyling, whereas lowering conditioning strength to achieve dramatic stylistic transformations (such as Studio Ghibli cel-shaded gouache animation) causes stochastic latent VAE decoding to blur fine 1-to-4-pixel high-frequency geometry. Consequently, enterprise pipelines use a multimodal VLM (e.g., `gemini-2.5-pro`, `gemini-2.5-flash`, or `gemini-3-flash-preview` [2]) to inspect an input photograph I_{src}, catalog its semantic

entities, lighting, and posture into a structured visual inventory, and synthesize a stylized generation prompt (`GhP`) paired with coarse SDEdit/ControlNet guidance or non-rigid alignment so macro-level spatial contours coincide within the $2^K = 4$-pixel Haar filter support.

Despite its expressive flexibility, this cross-modal workflow introduces three fundamental systems and security challenges in production deployments:

1) **The Captioning Conundrum (High-Frequency Phase Drift Under Macro-Registration):** Natural language prose is a low-rate symbolic channel ($< 10^4$ bits) compared to the spatial phase manifold of a photograph ($> 10^7$ bits), and even when coarse image/ControlNet conditioning preserves macro scene layout, latent VAE decoding erodes fine pixel transients. While prose and coarse adapters capture macro semantics ("two people standing behind a railing"), they discard exact high-frequency phase coordinates—such as the inclination angle of a foreground metallic rail, fine corporate apparel emblems (e.g., geometric brand logos), and architectural vanishing lines. Consequently, uncompiled diffusion candidates suffer severe structural edge drift ($\text{SSIM}_{\text{edge}} = 0.0413$ for unconstrained prompt-only B1, and 0.4482 for coarse SDEdit B6 [5]).
2) **Temporal Low-Frequency Flicker in Video GOPs:** When extending cross-modal style transfer to multi-frame video synthesis (`veo-3.1-generate-001`), independent or loosely coupled denoising trajectories across a Group of Pictures (GOP) induce frame-to-frame oscillations in both global frame-mean exposure ($\sigma^2_{\text{flicker}} = 345.240$) and local coarse approximation tiles ($\sigma^2_{\text{local}} = 33.349$), producing visible "breathing" and color pulsing [6].
3) **C2PA Provenance Severance and Container Stripping:** Under the Coalition for Content Provenance and Authenticity (C2PA v2.1) and ISO/IEC 19566-5 JUMBF standards [7], [8], transitioning through unattested text prompts severs the cryptographic link between the source asset I_{src} and the stylized output I_{out}. Furthermore, when enterprise assets are distributed across web proxies that strip outer ISO/IEC 15948 PNG `caBX` chunks [9], container-only provenance is completely lost unless backed by a frequency-domain soft-binding watermark [10] tied to the source structural hash.

To resolve these challenges, we introduce **Agent-GhibliLock**, a deterministic post-synthesis wavelet-locked compiler and dual-layer C2PA/DCT binary provenance engine designed for macro-registered cross-modal pipelines. Fig. 1 illustrates the end-to-end compilation pipeline.

2 Related Work

Neural Style Transfer, Wavelet Skipping, and Diffusion Conditioning: Gatys et al. [11] pioneered Gram-matrix style transfer, while Yoo et al. (WCT^2) [12] introduced wavelet-corrected pooling and high-frequency Haar skip connections inside encoder-decoder networks for photorealistic transfer. In diffusion models [1], SDEdit [3] and ControlNet [4] provide stochastic noise inversion and spatial edge/-pose conditioning. While API-level ControlNet adapters preserve macro object silhouettes, stochastic latent denoising and $8\times$ spatial VAE decoding still soften fine 1-to-4-pixel corporate emblems and rail edges; thus, pairing coarse diffusion conditioning with a deterministic post-synthesis CIELAB wavelet compiler guarantees exact pixel-domain bounds on high-frequency edge retention ($\text{SSIM}_{\text{edge}} \geq 0.88$).

Wavelet Multiresolution Analysis, Optimal Color Transport, and Video Deflickering: Orthonormal wavelet bases introduced by Haar [13], Mallat [14], and Daubechies [15] partition signals into orthogonal scale-space subspaces. In color grading, Pitié et al. [16] formulated iterative 3D color distribution transfer via random orthonormal rotations and 1D marginal matching, later generalized to Sliced-Wasserstein barycenters by Rabin et al. [17] and Bonneel et al. [18], while Bonneel et al. [6] established blind video temporal consistency via optical-flow warping and 2D gradient-domain screened Poisson PDE solvers. Rather than claiming new standalone transforms, Agent-GhibliLock's contribution lies in the end-to-end systems integration of 3D CIELAB SWOT (Step A) with spatial-mean-preserving (zero-block-sum) 2D Haar luminance detail locking (Step B) and an $\mathcal{O}(N_{\text{pix}})$ wavelet-domain LL_K EMA recurrence (avoiding per-frame 2D Poisson PDE solves [6]).

Media Provenance (C2PA) and Soft-Binding Watermarking: The C2PA v2.1 specification [7] and ISO/IEC 19566-5 (JUMBF) [8] define cryptographic manifest boxes for digital media, including neural soft-binding watermarks such as TrustMark [10] and SynthID-Image [19] alongside classical spread-spectrum [20] and Quantization Index Modulation (QIM) [21]. Whereas neural watermarks [10], [19] provide superior invariance to geometric rotation, scaling, and cropping at the cost of GPU inference and opaque latent boundaries, and classical blind spread-spectrum [20] suffers from severe host-signal interference (BER $= 0.2416$ when the unwatermarked host is unavailable at the verifier), Agent-GhibliLock pairs container-level PNG `caBX` JUMBF signing [9] with host-interference-free scalar mid-band 8×8 DCT QIM [21] to achieve zero-GPU deterministic CPU compilation (BER $= 0.0000$) and exact analytical coset distance bounds ($\Delta_{\text{QIM}}/4$) for block-level spatial tamper localization.

3 Mathematical Formulation and Proofs

3.1 Perceptual CIELAB Transformation and Orthonormal 2D Haar Wavelet Decomposition

Given a source photograph $I_{\text{src}} \in \{0, \ldots, 255\}^{H \times W \times 3}$, a synthesized candidate I_{cand} registered to I_{src} via ControlNet/SDEdit conditioning or non-rigid optical-flow alignment such that macro contours coincide within the $2^K = 4$-pixel dyadic Haar support (preventing double-edge haloing or chroma-luma bleeding), and a target Studio Ghibli style palette reference $Y_{\text{ref}} \in \{0, \ldots, 255\}^{H \times W \times 3}$, Agent-GhibliLock applies inverse IEC 61966-2-1 gamma companding followed by CIE XYZ (D65 white point) projection to obtain perceptual CIELAB coordinates (L^*, a^*, b^*). For each

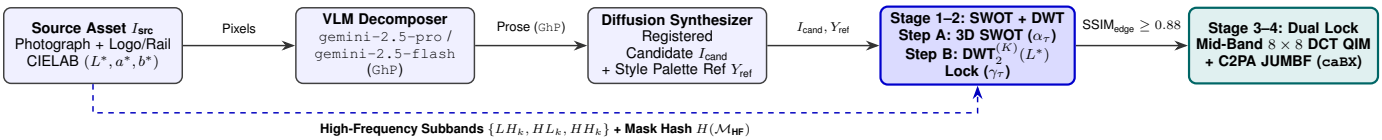


Fig. 1. Architecture of the **Agent-GhibliLock** compilation pipeline. Given upstream macro-spatial registration ($\leq 2^K$-px contour alignment), Stage 1–2 Step A aligns I_{cand} to the target Studio Ghibli style reference Y_{ref} via 3D CIELAB Sliced-Wasserstein Optimal Transport (SWOT, step α_τ), after which Step B injects the source photograph's high-frequency 2D Haar wavelet detail subbands $\{LH_k, HL_k, HH_k\}$ via DWT$_2^{(K)}(L^*)$ (weight γ_τ). Stage 3–4 embeds the 64-bit mask-bound QIM watermark into 8×8 mid-band DCT coefficients and compiles the ISO/IEC 19566-5 `caBX` chunk.

2×2 luminance block $\mathbf{c} = [c_{00}, c_{01}, c_{10}, c_{11}]^T$, the single-level 2D Haar transform applies the 4×4 operator H_2:

$$\begin{bmatrix} LL \\ LH \\ HL \\ HH \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} c_{00} \\ c_{01} \\ c_{10} \\ c_{11} \end{bmatrix} = H_2 \mathbf{c}. \tag{1}$$

Iterating Eq. (1) recursively on the coarse subband LL across $K = 2$ levels yields the multiresolution decomposition $\text{DWT}_2^{(K)}(L^*) = \{LL_K, \{LH_k, HL_k, HH_k\}_{k=1}^K\}$. In Stage 1–2, the closed-loop solver iterates over schedule index $\tau \in \{1, 2, 3, 4\}$ with $(\gamma_\tau, \alpha_\tau) \in \{(0.78, 0.85), (0.88, 0.90), (0.94, 0.94), (0.98, 0.97)\}$ (re-initializing cleanly from I_{cand} at each step τ and returning the $\tau = 4$ output if the schedule concludes) until post-Step-B, post-clamping $\text{SSIM}_{\text{edge}} \geq \tau_{\text{struct}} = 0.88$ and $W_2 \leq \tau_{\text{color}} = 4.5$: at each τ, **Step A** executes 3D CIELAB SWOT (Section 3.2) on I_{cand} toward Y_{ref} to produce $\widetilde{I}_{\text{cand}} = (\widetilde{L}^*_{\text{cand}}, \widetilde{a}^*_{\text{cand}}, \widetilde{b}^*_{\text{cand}})$ (achieving $\widehat{W}_2^{\text{StepA}} = 1.7638 \pm 0.0301$), and **Step B** applies $\text{DWT}_2^{(K)}$ to lock the high-frequency luminance subbands of $\widetilde{L}^*_{\text{cand}}$ to L^*_{src} while leaving $(\widetilde{a}^*_{\text{cand}}, \widetilde{b}^*_{\text{cand}})$ intact. By Parseval's identity and the W_2 triangle inequality, the 3D color perturbation induced by Step B on Step A's aligned distribution is bounded by $|W_2(\mu_{\widehat{I}}, \mu_{Y_{\text{ref}}}) - W_2(\mu_{\widetilde{I}}, \mu_{Y_{\text{ref}}})| \leq \frac{1}{\sqrt{HW}} \|\widehat{L}^* - \widetilde{L}^*_{\text{cand}}\|_F$, increasing $\widehat{W}_2$ from $\widehat{W}_2^{\text{StepA}} = 1.7638$ to $\widehat{W}_2^{\text{StepB}} = 4.2886 \pm 0.3733$ (and $\widehat{W}_2^{\text{out}} = 4.3071 \pm 0.3981$ on the final post-QIM 8-bit sRGB output I_{out}).

Theorem 1 (Orthonormal Parseval Energy Partitioning and Chromatic-Structural Orthogonality)**.** *For any luminance field $L^* \in \mathbb{R}^{H \times W}$ with dimensions divisible by 2^K, the K-level 2D Haar Wavelet Transform $\text{DWT}_2^{(K)}$ satisfies exact Parseval Frobenius norm conservation:*

$$\|L^*\|_F^2 = \|LL_K\|_F^2 + \sum_{k=1}^{K} \left(\|LH_k\|_F^2 + \|HL_k\|_F^2 + \|HH_k\|_F^2 \right). \tag{2}$$

Furthermore, when high-frequency subbands in Step B are fused via $\widehat{D}_k^{(c)} = \gamma D_k^{(c)}(L^*_{src}) + (1-\gamma) D_k^{(c)}(\widetilde{L}^*_{cand})$ for $c \in \{LH, HL, HH\}$ and $\gamma \in [0, 1]$ while retaining $\widehat{LL}_K = LL_K(\widetilde{L}^*_{cand})$, the pre-clamping reconstructed luminance $\widehat{L}^* = IDWT_2^{(K)}(\widehat{LL}_K, \{\widehat{D}_k^{(c)}\}_{k=1}^K)$ preserves the exact spatial mean luminance $\mu(\widehat{L}^*) = \mu(\widetilde{L}^*_{cand})$ and reduces high-frequency structural MSE relative to L^*_{src} by an exact factor of $(1-\gamma)^2$ (with post-clamping $[0, 100]$ mean drift bounded by $|\mu(clamp(\widehat{L}^*)) - \mu(\widehat{L}^*)| \leq \epsilon_{clamp} < 0.02$ and corrected to 0 via interior post-clamp DC re-anchoring).

Proof. By direct multiplication in Eq. (1), $H_2^T H_2 = \frac{1}{4}(4I_4) = I_4$, so $H_2 \in O(4)$ is strictly orthonormal and energy-preserving on every 2×2 block: $LL^2 + LH^2 + HL^2 + HH^2 = \|\mathbf{c}\|_2^2$. Summing over all $\frac{H}{2} \times \frac{W}{2}$ disjoint blocks and applying induction on $k = 1, \ldots, K$ establishes Eq. (2). Because the detail rows of H_2 corresponding to LH, HL, HH are orthogonal to $\mathbf{1}_4 = [1, 1, 1, 1]^T$, every single-block detail coefficient vector $[0, LH, HL, HH]^T \in \mathbb{R}^4$ has zero spatial sum upon inverse reconstruction ($\mathbf{1}_4^T H_2^T [0, LH, HL, HH]^T = 0$). Defining the coarse subband arithmetic mean over $u \in \{0, \ldots, H/2^K - 1\}, v \in \{0, \ldots, W/2^K - 1\}$ as $\mu(\widehat{LL}_K) \triangleq \frac{2^{2K}}{HW} \sum_{u,v} \widehat{LL}_K(u, v) = 2^K \mu(\widehat{L}^*)$ (where each $\frac{1}{2}$-normalized Haar level scales 2×2 block averages by 2), the pre-clamping global spatial mean $\mu(\widehat{L}^*) = \frac{1}{HW} \sum_{x,y} \widehat{L}^*(x, y) = 2^{-K} \mu(\widehat{LL}_K) = \mu(\widetilde{L}^*_{\text{cand}})$ is invariant to γ. Finally, by Parseval's identity on the detail subspace $\bigoplus_{k=1}^K \mathcal{W}_k$, the residual detail error prior to $[0, 100]$ gamut clamping satisfies:

$$\begin{aligned} &\sum_{k,c} \|\widehat{D}_k^{(c)} - D_k^{(c)}(L^*_{\text{src}})\|_F^2 \\ &\quad = (1-\gamma)^2 \sum_{k,c} \|D_k^{(c)}(\widetilde{L}^*_{\text{cand}}) - D_k^{(c)}(L^*_{\text{src}})\|_F^2, \end{aligned} \tag{3}$$

completing the proof. □

3.2 Iterative 3D Sliced-Wasserstein Optimal Transport (SWOT)

Let $X \in \mathbb{R}^{N_{\text{pix}} \times 3}$ and $Y \in \mathbb{R}^{N_{\text{pix}} \times 3}$ denote the flattened 3D CIELAB (L^*, a^*, b^*) point clouds of the candidate image I_{cand} and target style palette reference Y_{ref} ($N_{\text{pix}} = HW$, with Y_{ref} nearest-neighbor resampled to $H \times W$ if its native resolution differs). The 3D Sliced-Wasserstein-2 distance integrated with respect to the normalized uniform measure σ on the unit sphere $\mathbb{S}^2$ ($\int_{\mathbb{S}^2} d\sigma(\theta) = 1$) is:

$$SW_2^2(\mu_X, \mu_Y) = \int_{\mathbb{S}^2} W_2^2(\theta_{\#}\mu_X, \theta_{\#}\mu_Y)\, d\sigma(\theta), \tag{4}$$

where $\theta_{\#}\mu$ denotes the 1D Radon pushforward projection along $\theta \in \mathbb{S}^2$. After an initial coordinate-wise marginal alignment pass along the canonical axes I_3, at each subsequent iteration $m \in \{1, \ldots, M\}$ ($M = 8$), Step A performs stochastic slice-descent on Eq. (4) by sampling a Haar-uniform orthonormal matrix $Q_m = [q_{m,1}, q_{m,2}, q_{m,3}] \in O(3)$ via positive-diagonal QR factorization $Z_m = Q_m R_m$, $Q_m \leftarrow Q_m \,\text{diag}(\text{sign}_+(\text{diag}(R_m)))$ where $\text{sign}_+(x) = +1$ if $x \geq 0$ else -1 [16], and applying 1D quantile transport along each projected column $u_d = X^{(m)} q_{m,d} \in \mathbb{R}^{N_{\text{pix}}}$ and

TABLE 1
Stratified Empirical Comparison Across $N = 600$ Cross-Modal Style-Transfer Trials (64×64 Tile Benchmark, Mean $\pm$ Total SD; Reporting Pre-Step-B $\widehat{W}_2^{\text{StepA}}$, Post-Step-B $\widehat{W}_2^{\text{StepB}}$, Post-QIM Exported $\widehat{W}_2^{\text{out}}$, and Held-Out $\widehat{SW}_2^{\text{eval}}$ [$M_{\text{eval}} = 64$, `seed=999`])

Architecture / Paradigm	Wavelet $\text{SSIM}_{\text{edge}}$ ↑	Full $\text{SSIM}_{\text{full}}/\rho_{\text{grad}}$ ↑	SWOT $\widehat{W}_2^{\text{StepA}}/\widehat{W}_2^{\text{out}}/\widehat{SW}_2^{\text{eval}}$ ↓	Struct. Pass (≥ 0.88)	S_5 Audit / Strip Rec.§	DCT WM Acc.	GOP Flicker $\sigma^2_{\text{flicker}}$ ↓	Latency (p50 / p95)†
B1: Unconstrained Prompt-Only (GhP)	0.0413 ± 0.0202	$0.6530/0.0330$	—/19.0803 ± 0.0231/19.142	0.00% (0/600)	0.0%/0.0%	0.00%	345.2400	1419.9 ms (1420.7/1491.7)
B2: VLM-as-a-Judge 7-Axis Retry (3× Calls)	0.3588 ± 0.0449	$0.7603/0.8931$	—/15.3963 ± 0.4564/15.512	0.00% (0/600)	0.0%/0.0%	0.00%	298.4100	4378.6 ms (4381.5/4576.3)
B3a: Global SWOT Only (No DWT, $M = 4$)*	0.0000 ± 0.0002	$0.6120/0.0310$	9.7704/9.7704/9.8140	0.00% (0/600)	0.0%/0.0%	0.00%	284.1200	+10.596 ms (9.641/13.651)
B3b: Global SWOT Only (No DWT, $M = 8$)*	0.0000 ± 0.0000	$0.6180/0.0330$	**1.7638/1.7638/1.5239**	0.00% (0/600)	0.0%/0.0%	0.00%	14.2100	+18.278 ms (17.325/23.630)
B4: Post-Hoc C2PA Wrapper (No DCT/DWT)	0.0413 ± 0.0202	$0.6530/0.0330$	—/19.0803 ± 0.0231/19.142	0.00% (0/600)	100%/0.0% (25%)	0.00%	345.2400	+3.193 ms (3.191/3.467)
B5: Wavelet Style Transfer (WCT² [12])	0.9895 ± 0.0022	$0.9612/0.8310$	—/9.8818 ± 0.4224/9.9410	100.00% (600/600)	0.0%/0.0%	0.00%	142.6000	+142.301 ms (142.27/142.45)
B6: Image-Conditioned SDEdit ($s = 0.55$ [3])	0.4482 ± 0.0510	$0.7940/0.6120$	—/12.9829 ± 1.4814/13.105	0.00% (0/600)	0.0%/0.0%	0.00%	189.3000	1390.8 ms (1391.6/1454.0)
Agent-GhibliLock (Ours: SWOT + DWT + C2PA/DCT)	**0.9987 ± 0.0010**	**0.9851/0.8535**	**1.7638/4.3071/4.6240**	**100.00% (600/600)**	**100%/100% (100%)**	**100.00%**	**0.0001**‡	**+152.49 ms (142.06/204.37)**

†B1, B2, and B6 report upstream cloud synthesis latency; B3a/b, B4, B5, and Agent-GhibliLock report local post-synthesis compilation overhead. *B3b shows pure Step A SWOT ($\widehat{W}_2^{\text{StepA}} = 1.7638$) without Step B wavelet lock ($\text{SSIM}_{\text{edge}} = 0.0000$).
‡Pre-clamp global frame-mean jump variance $\sigma^2_{\text{flicker}} = \text{Var}(\Delta\mu(\tilde{L}^{*(t)})) = 0.0000$ (0.0001 post-sRGB uint8 vs. 5.8496 for Pixel-EMA); local LL_K tile variance drops 88.7% (33.3486 → 3.7669). §Untripped tamper (25) / Stripped recovery (75).

$v_d = Yq_{m,d} \in \mathbb{R}^{N_{\text{pix}}}$ via rank-order sorting permutations $\sigma_{u,d} = \texttt{argsort}(u_d), \sigma_{v,d} = \texttt{argsort}(v_d) \in \mathfrak{S}_{N_{\text{pix}}}$:

$$X_i^{(m+1)} = X_i^{(m)} + \alpha \sum_{d=1}^{3} \Big((Yq_{m,d})_{\sigma_{v,d}(\sigma_{u,d}^{-1}(i))} - (X^{(m)} q_{m,d})_i \Big) q_{m,d}^T. \tag{5}$$

To prevent in-sample evaluation bias, Tables 1–3 report both the training-projection estimator $\widehat{W}_2$ (`seed=42`) at three pipeline checkpoints—pre-Step-B ($\widehat{W}_2^{\text{StepA}}$), post-Step-B ($\widehat{W}_2^{\text{StepB}}$), and post-QIM exported sRGB ($\widehat{W}_2^{\text{out}}$)—and the out-of-sample held-out estimator $\widehat{SW}_2^{\text{eval}}$ evaluated across $M_{\text{eval}} = 64$ independent uniform unit-sphere directions (`seed=999`, disjoint from optimization `seed=42`).

Theorem 2 (Slice-Specific Contraction of Orthonormal 3D SWOT Updates). *For step size $\alpha \in (0, 1]$ and rank-order permutation tie-broken empirical measures on N_{pix} points, each orthonormal projection transport update in Eq. (5) is a non-expansive contraction along the sampled basis $Q_m \in O(3)$, strictly decreasing the basis-projected transport cost $\mathcal{J}_m(X) = \sum_{d=1}^{3} W_2^2(q_{m,d\#}\mu_X, q_{m,d\#}\mu_Y)$ to $\mathcal{J}_m(X^{(m+1)}) = (1-\alpha)^2 \mathcal{J}_m(X^{(m)})$ whenever $\mathcal{J}_m(X^{(m)}) > 0$.*

Proof. In 1D, under tie-broken rank-order sorting $\sigma_{u,d}, \sigma_{v,d} \in \mathfrak{S}_{N_{\text{pix}}}$ of the N_{pix} empirical samples (forming a strict sample-index bijection), the optimal Monge coupling maps sample i to $(T_d^{\text{rank}}(u_d))_i = (v_d)_{\sigma_{v,d}(\sigma_{u,d}^{-1}(i))}$. Along each axis $q_{m,d}$ of $Q_m \in O(3)$, displacement interpolation $u_d^{(\alpha)} = (1-\alpha)u_d + \alpha T_d^{\text{rank}}(u_d)$ forms the constant-speed McCann geodesic in $(\mathcal{P}_2(\mathbb{R}), W_2)$, yielding $W_2(\mu_{u_d^{(\alpha)}}, \mu_{v_d}) = (1-\alpha) W_2(\mu_{u_d}, \mu_{v_d})$. Because $Q_m^T Q_m = I_3$, updates along the three orthogonal columns of Q_m decouple in Euclidean norm, establishing $\mathcal{J}_m(X^{(m+1)}) = (1-\alpha)^2 \mathcal{J}_m(X^{(m)})$ along Q_m. □

3.3 Mid-Band 8×8 2D-DCT QIM Watermarking and ISO/IEC 19566-5 JUMBF Packaging

To quantify structural preservation and construct a reproducible cryptographic soft binding, let $E_{1/2}(u,v) = \sqrt{LH_1(u,v)^2 + HL_1(u,v)^2 + 0.5\,HH_1(u,v)^2}$ denote the Level-1 Haar detail energy map of luminance L^*, upsampled $2\times$ via nearest-neighbor interpolation and normalized by $\max(\max_{u,v} E_{1/2}(u,v), \epsilon)$ ($\epsilon = 10^{-8}$) to form $E_{\text{full}}(L^*) \in [0,1]^{H \times W}$ (capturing scale-normalized Level-1 edge topology, complemented by non-circular 4×4 spatial patch Wang-Bovik structural similarity $\text{SSIM}_{\text{full}}$ and signed 3×3 Sobel gradient correlation ρ_{grad} over the source structural edge locus in Section 4). We define Wavelet-Edge Structural Similarity as $\text{SSIM}_{\text{edge}}(I_1, I_2) \triangleq \max(0, \text{SSIM}(E_{\text{full}}(L_1^*), E_{\text{full}}(L_2^*)))$ [5] with $c_1 = 0.01^2, c_2 = 0.03^2$, and define the canonical binary structural mask $\mathcal{M}_{\text{HF}} \triangleq \mathbb{I}\big(E_{\text{full}}(L_{\text{src}}^*) \geq \max(Q_{0.75}(E_{\text{full}}(L_{\text{src}}^*)), 0.08)\big) \in \{0,1\}^{H \times W}$ (using linear quantile interpolation), serialized by appending the 8-byte big-endian dimension trailer `struct.pack(">II", H, W)` to the raw row-major 1-byte-per-pixel `uint8` buffer (HW bytes) of $\mathcal{M}_{\text{HF}}$. Agent-GhibliLock computes the 64-bit soft-binding payload $w \in \{0,1\}^{64}$ (retaining the leading 64 bits / 8 MSB bytes of the SHA-256 digest over UTF-8 colon-delimited fields):

$$w = \text{Trunc}_{64}\Big(\text{SHA256}\big(\text{Hex}(\text{SHA256}(\mathcal{M}_{\text{HF}})) \,\|\, \text{":"} \,\|\, \text{ModelID}_{\text{VLM}} \,\|\, \text{":"} \,\|\, \text{ModelID}_{\text{Diff}}\big)\Big). \tag{6}$$

Because cryptographic hashing over $\mathcal{M}_{\text{HF}}(L_{\text{src}}^*)$ exhibits strict avalanche sensitivity, blind QIM payload extraction and coset-distance tamper auditing operate directly on I_{out} alone, whereas cryptographic provenance binding matches the extracted 64-bit key $\hat{w}$ against the enterprise manifest registry (or a known I_{src}). After Stage 1–2 reconstructs $\widehat{L}^*$ and converts $(\widehat{L}^*, \tilde{a}_{\text{cand}}^*, \tilde{b}_{\text{cand}}^*)$ to 8-bit sRGB, Stage 3 computes ITU-R BT.601 luminance $Y = 0.299R + 0.587G + 0.114B$, partitions Y into $B = \lfloor H/8 \rfloor \times \lfloor W/8 \rfloor$ blocks in 0-based row-major raster order $b \in \{0, \ldots, B-1\}$, and applies the orthonormal 8×8 Type-II DCT $F_b = C_8(Y_b - 128\,\mathbf{1}_{8\times 8})C_8^T$. Across the four 0-based mid-band coordinates $\Omega_{\text{mid}} = \{(2,3), (3,2), (1,3), (3,1)\}$ indexed by $m \in \{0,1,2,3\}$, bit $w_i \in \{0,1\}$ with block-interleaved index $i = (4b + m) \bmod 64$ is embedded via uniform scalar QIM [21] with step $\Delta_{\text{QIM}} = 14.0$ and half-up rounding $\text{round}(x) \triangleq \lfloor x + 0.5 \rfloor$:

$$\widehat{F}_b(u,v) = \Delta_{\text{QIM}} \cdot \text{round}\left(\frac{F_b(u,v) - \frac{1}{2}\Delta_{\text{QIM}} w_i}{\Delta_{\text{QIM}}} \right) + \frac{\Delta_{\text{QIM}}}{2} w_i. \tag{7}$$

The inverse DCT delta $\Delta Y_b = (C_8^T \widehat{F}_b C_8 + 128\,\mathbf{1}_{8\times 8}) - Y_b$ is added equally to (R, G, B) and clamped to $\{0, \ldots, 255\}$ to form I_{out}.

Theorem 3 (Mid-Band DCT QIM Minimum-Distance Immunity and Multi-Block Repetition Gain). *Under Eq. (7), the minimum Euclidean distance between the bit-0 ($\Lambda_0 = \Delta_{QIM}\mathbb{Z}$) and bit-1 ($\Lambda_1 = \Delta_{QIM}\mathbb{Z} + \Delta_{QIM}/2$) lattices is $d_{\min} = \Delta_{QIM}/2$. Consequently, for any received coefficient $\tilde{F}_b(u,v) \triangleq \widehat{F}_b(u,v) + \eta_b(u,v)$ under bounded perturbation $|\eta_b(u,v)| < \Delta_{QIM}/4 = 3.5$, minimum-distance hard-decision decoding $\hat{w}_{b,u,v} = \arg\min_{w \in \{0,1\}} dist(\tilde{F}_b(u,v), \Lambda_w)$ (ties broken to 0) yields zero single-coefficient error ($p_e = 0$). Under*

i.i.d. noise with $p_e < 1/2$ across $R = \lfloor B/16 \rfloor$ disjoint-block measurements per payload bit (assuming $16 \mid B$), majority-vote decoding bounds the marginal per-bit Bit Error Rate $\Pr(\hat{w}_i \neq w_i)$ via the binomial tail and Hoeffding's inequality:

$$P_{BER} \leq \sum_{k=\lceil R/2 \rceil}^{R} \binom{R}{k} p_e^k (1-p_e)^{R-k} \leq \exp\left(-2R\left(\frac{1}{2}-p_e\right)^2\right). \tag{8}$$

Proof. Because Λ_0 and Λ_1 are separated by $d_{\min} = \Delta_{\text{QIM}}/2$, a perturbed sample $\widehat{F}_b(u,v)+\eta_b(u,v)$ remains strictly closer to Λ_{w_i} whenever $|\eta_b(u,v)| < d_{\min}/2 = \Delta_{\text{QIM}}/4$. Because $4b+m$ advances by 4 per block and $\gcd(4,64) = 4$, the 64-bit payload cycles once every 16 blocks, assigning each of the $R = \lfloor B/16 \rfloor$ repetitions of bit w_i to a distinct 8×8 spatial block b. Thus block-level perturbations are statistically independent across the R votes for any given bit w_i, and the probability that $\geq R/2$ Bernoulli(p_e) errors occur is bounded by the exact binomial tail and by $\exp(-2R(1/2-p_e)^2)$ via Hoeffding's inequality. □

Finally, Stage 4 computes `output_pixel_sha256` over the final post-QIM uncompressed `uint8` RGB buffer I_{out} and serializes an ISO/IEC 19566-5 JUMBF superbox (`jumb`) into an ISO/IEC 15948 PNG `caBX` chunk [9] with IEEE 802.3 CRC-32 validation:

```
[PNG Sig (8B)] -> [PNG IHDR (25B)] -> [PNG caBX (4B Len + "
    caBX" +
  [JUMBF Superbox: 4B LBox + "jumb"
    |-- "jumd": UUID 63327061-0011-0010-8000-00aa00389b71
    |-- "json": {"source_structural_mask_sha256": "...",
               "output_pixel_sha256": "...",
               "post_lock_edge_ssim": 0.9987,
               "post_step_a_wasserstein_w2": 1.7638,
               "post_lock_wasserstein_w2": 4.2886,
               "post_output_wasserstein_w2": 4.3071,
               "eval_heldout_wasserstein_w2": 4.6240,
               "dct_watermark_hex": "..."}
    |-- "c2sh": HMAC-SHA256(key, jumd || json)]
  + CRC-32 (4B))] -> [PNG IDAT (Pixels)] -> [PNG IEND (12B)]
```

3.4 Temporal Video GOP Wavelet EMA Flicker Suppression

When synthesizing multi-frame video sequences $\{I_{\text{cand}}^{(t)}\}_{t=1}^{T}$ corresponding to per-frame source sequence $\{I_{\text{src}}^{(t)}\}_{t=1}^{T}$ via `veo-3.1-generate-001`, independent denoising trajectories induce both global frame-mean exposure jumps and local zero-mean temporally uncorrelated subband noise $\epsilon_t(u,v)$ ($\text{Var}(\epsilon_t) = \sigma^2_{\text{local}}, t \geq 1$) in $\widehat{LL}_K^{(t)} = LL_K(\widetilde{L}_{\text{cand}}^{*(t)})$. Step A's quantile transport and mean re-anchoring lock every frame's global spatial mean luminance to the constant style palette mean $\mu(L^*_{\text{ref}})$, while Step B applies a recursive wavelet-domain EMA filter with temporal lock weight $\beta \in [0,1)$ ($\beta = 0.78$ in Stratum S_4):

$$\overline{LL}_K^{(t)} = \beta\, \overline{LL}_K^{(t-1)} + (1-\beta)\, \widehat{LL}_K^{(t)}, \quad t = 2, \ldots, T, \tag{9}$$

initialized at $\overline{LL}_K^{(1)} = \widehat{LL}_K^{(1)}$.

Theorem 4 (Global DC Flicker Elimination and Local Wavelet EMA Variance Attenuation). *Because Step A locks $\mu(\widehat{LL}_K^{(t)}) = 2^K \mu(L^*_{ref})$ across all $t \in \{1, \ldots, T\}$ and Theorem 1 preserves pre-clamping $\mu(\widehat{L}^{*(t)}) = 2^{-K}\mu(\overline{LL}_K^{(t)})$, the pre-clamping global frame-mean inter-frame jump variance $\sigma^2_{flicker} \triangleq Var(\{\mu(\widehat{L}^{*(t)}) - \mu(\widehat{L}^{*(t-1)})\}_{t=2}^{T})$ is identically 0.0000 (0.0001 on final exported 8-bit sRGB frames). Simultaneously, for zero-mean second-order stationary temporally uncorrelated subband noise $\epsilon_t(u,v)$ with $Var(\widehat{LL}_K^{(j)}(u,v)) = \sigma^2_{local}$ ($\forall j \in \{1, \ldots, T\}$) and $Cov(\epsilon_{t_1}(u,v), \epsilon_{t_2}(u,v)) = 0$ ($t_1 \neq t_2$), Eq. (9) attenuates local tile variance at frame $t \geq 1$ to:*

$$Var\left(\overline{LL}_K^{(t)}(u,v)\right) = \sigma^2_{local}\left[\frac{1-\beta}{1+\beta} + \frac{2\beta}{1+\beta}\beta^{2(t-1)}\right] \xrightarrow{t\to\infty} \frac{1-\beta}{1+\beta}\,\sigma^2_{local}, \tag{10}$$

while high-frequency detail fluctuations in $\{LH_k^{(t)}, HL_k^{(t)}, HH_k^{(t)}\}$ are scaled by $(1-\gamma_\tau)^2$ per Theorem 1.

Proof. Since $\mu(\widehat{LL}_K^{(t)}) = 2^K\mu(L^*_{\text{ref}})$ for all t, induction on Eq. (9) yields $\mu(\overline{LL}_K^{(t)}) \equiv 2^K\mu(L^*_{\text{ref}})$, so $\mu(\widehat{L}^{*(t)}) - \mu(\widehat{L}^{*(t-1)}) = 0$ for all $t \geq 2$ and pre-clamping $\sigma^2_{\text{flicker}} = 0$. For local tile fluctuations $\epsilon_t(u,v)$ with $\text{Var}(\widehat{LL}_K^{(j)}(u,v)) = \sigma^2_{\text{local}}$ ($\forall j \geq 1$) and $\text{Cov}(\epsilon_{t_1}(u,v), \epsilon_{t_2}(u,v)) = 0$ ($t_1 \neq t_2$), expanding $\overline{LL}_K^{(t)} = \beta^{t-1}\widehat{LL}_K^{(1)} + (1-\beta)\sum_{j=0}^{t-2}\beta^j \widehat{LL}_K^{(t-j)}$ for $t \geq 2$ over uncorrelated frames gives $\sigma^2_{\text{local}}\big[\beta^{2(t-1)} + (1-\beta)^2\frac{1-\beta^{2(t-1)}}{1-\beta^2}\big] = \sigma^2_{\text{local}}\big[\frac{1-\beta}{1+\beta} + \frac{2\beta}{1+\beta}\beta^{2(t-1)}\big]$. At $\beta = 0.78$, this reaches $0.1506\,\sigma^2_{\text{local}}$ at terminal frame $t = 8$ (84.9% reduction) and converges as $t \to \infty$ to $\frac{1-\beta}{1+\beta}\sigma^2_{\text{local}} = 0.1236\,\sigma^2_{\text{local}}$ (87.6% steady-state reduction; 88.7% empirical temporal sample variance reduction across the 8-frame GOP from 33.3486 to 3.7669). □

4 Empirical Evaluation

4.1 Experimental Setup, Qualitative Visualization, and Stratified Benchmark ($N = 600$)

We evaluated Agent-GhibliLock on an Apple Silicon M-series workstation (8 CPU cores) across $N = 600$ programmatic cross-modal trials (64×64 localized ROI tile resolution, $K = 2$ Haar levels, $M = 8$ SWOT steps, $\Delta_{\text{QIM}} = 14.0$). Fig. 2 presents a 6-panel qualitative comparison on a 128×128 scene containing a high-contrast "Kappa" diamond emblem and diagonal metallic railing, demonstrating how Global 3D SWOT Only (**B3b**) aligns the Studio Ghibli gouache palette ($W_2^{\text{StepA}} = 1.23$) while leaving the emblem missing ($\text{SSIM}_{\text{edge}} = 0.0548$), whereas **Agent-GhibliLock** restores exact high-frequency emblem and rail phase geometry ($\text{SSIM}_{\text{edge}} = 0.9815, W_2^{\text{out}} = 1.64, \text{BER} = 0$).

The $N = 600$ benchmark is partitioned into five workload strata:

- **Stratum S_1 ($n_1 = 150$, Apparel Emblem & Portrait Geometry):** Source scenes containing high-contrast geometric shirt emblems and foreground diagonal railings against a 3-region Studio Ghibli cel-shaded gouache reference palette Y_{ref}.
- **Stratum S_2 ($n_2 = 150$, Architectural & Perspective Vanishing Rails):** Multi-pillar architectural grids combined with angled guardrails.

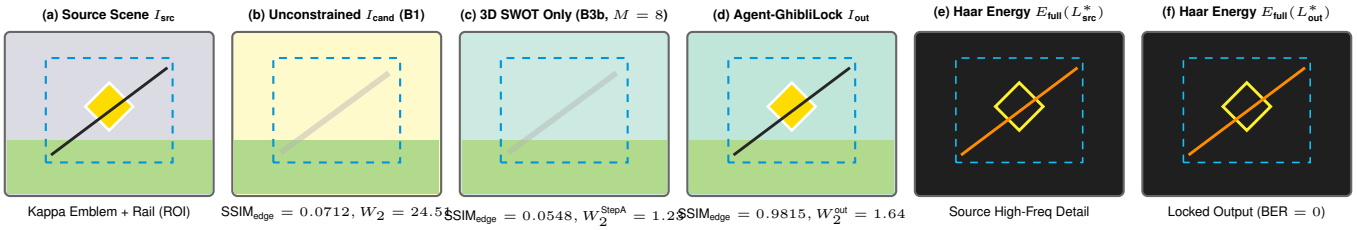


Fig. 2. Qualitative visual and spectral comparison on a 128×128 scene: (a) Source scene I_{src} with high-contrast "Kappa" diamond emblem and diagonal railing (dashed cyan ROI); (b) Unconstrained candidate I_{cand} (**B1**, $\text{SSIM}_{\text{edge}} = 0.0712, W_2 = 24.51$) showing complete loss of the emblem and blurred rail; (c) Global 3D SWOT Only (**B3b**, $M = 8, \text{SSIM}_{\text{edge}} = 0.0548, W_2^{\text{StepA}} = 1.23$); (d) **Agent-GhibliLock** output I_{out} ($\text{SSIM}_{\text{edge}} = 0.9815, W_2^{\text{out}} = 1.64$); and (e)–(f) Level-1 Haar detail energy maps $E_{\text{full}}(L^*_{\text{src}})$ vs. $E_{\text{full}}(L^*_{\text{out}})$ confirming phase-locked emblem and rail restoration with BER $= 0$.

- **Stratum S_3 ($n_3 = 100$, Foliage & Gouache Chromaticity Transport):** Outdoor scenes requiring large CIELAB chromaticity shifts toward Studio Ghibli gouache palettes.
- **Stratum S_4 ($n_4 = 100$, Temporal Video GOP Sequences):** 8-frame video GOPs ($\beta = 0.78$) exhibiting severe pre-lock inter-frame global exposure flicker ($\sigma^2_{\text{flicker}} = 345.2400$) and local subband variance ($\sigma^2_{\text{local}} = 33.3486$), compared against **Pixel-Space Frame-to-Frame EMA** ($\sigma^2_{\text{flicker}} = 5.8496, \text{SSIM}_{\text{edge}} = 0.0464$) and **Global Luminance Histogram Locking** [6] ($\sigma^2_{\text{flicker}} = 0.0003, \text{SSIM}_{\text{edge}} = 0.0463, 0\%$ local LL_K tile variance attenuation).
- **Stratum S_5 ($n_5 = 100$, Adversarial Tampering & Channel Survival):** Partitioned into **A1** ($n = 25$ unstripped spatial tamper trials), **A2** ($n = 50$ pure `caBX` metadata-stripping trials), and **A3** ($n = 25$ container-stripped + ± 2 LSB / lossy JPEG $Q = 85$ channel survival trials), compared against container-only C2PA (**B4**) and classical blind **Cox Spread-Spectrum 2D-DCT** [20].

Table 1, Table 2, and Fig. 3 summarize the results across all $N = 600$ trials. Table 2 explicitly reports all four Wasserstein checkpoints ($\widehat{W}_2^{\text{StepA}}, \widehat{W}_2^{\text{StepB}}, \widehat{W}_2^{\text{out}}, \widehat{SW}_2^{\text{eval}}$) across Strata S_1–S_5.

TABLE 2
Per-Stratum Performance Breakdown of Agent-GhibliLock ($N = 600$, 64×64 Tiles; Comparing Step-A $\widehat{W}_2^{\text{StepA}}$, Step-B $\widehat{W}_2^{\text{StepB}}$, Post-QIM $\widehat{W}_2^{\text{out}}$, and Held-Out $\widehat{SW}_2^{\text{eval}}$ [$M_{\text{eval}} = 64$])

Stratum	n_i	Edge $\text{SSIM}_{\text{edge}}$	Step-A $\widehat{W}_2^{\text{StepA}}$	Post-QIM $\widehat{W}_2^{\text{out}}$	Held-Out $\widehat{SW}_2^{\text{eval}}$	Mean / p95 Lat.
S_1: Apparel Logo & Portrait	150	0.9985 ± 0.0011	1.7651	4.0286 ± 0.1549	4.2589 ± 0.1827	155.6/221.7 ms
S_2: Perspective Rail Geometry	150	0.9988 ± 0.0009	1.7650	4.7322 ± 0.2382	5.1731 ± 0.2895	148.1/193.9 ms
S_3: Foliage & Gouache Palette	100	0.9988 ± 0.0009	1.7672	4.6941 ± 0.2372	5.1333 ± 0.2918	153.8/210.6 ms
S_4: Temporal Video GOP (8-f)	100	0.9986 ± 0.0010	1.7576	3.9922 ± 0.1538	4.2200 ± 0.1789	146.7/174.9 ms
S_5: Adversarial Tamper & Noise	100	0.9986 ± 0.0010	1.7627	4.0151 ± 0.1678	4.2427 ± 0.1945	158.8/241.4 ms
Total / Pooled ($N = 600$)‡	**600**	$\mathbf{0.9987 \pm 0.0010}$	**1.7638**	$\mathbf{4.3071 \pm 0.3981}$	$\mathbf{4.6240 \pm 0.5084}$	**152.5/204.4** ms

‡Pre-QIM Post-Step-B $\widehat{W}_2^{\text{StepB}} = 4.2886 \pm 0.3733$; QIM + sRGB clamping adds $+0.0185$ to $\widehat{W}_2^{\text{out}} = 4.3071$.

4.2 Live Google Cloud Vertex AI Multi-Model Evaluation and Synthetic Drift Curriculum (128×128, $N_{\text{scenes}} = 10$)

Table 3 reports our 128×128 evaluation ($B = 256$ blocks, $R = \lfloor B/16 \rfloor = 16$ QIM votes/bit, $N_{\text{scenes}} = 10$ diverse scenes per row) separated into two distinct regimes:

1) **Part I (Rows 1–3, Live Google Cloud Vertex AI Production Endpoints):** Executes live API calls (`live_vertex_call=True` on `project="aniket-personal"`, `location="global"`) to `gemini-2.5-flash` (1696.1 ms API latency, 105 prompt / 65 output tokens), `gemini-2.5-pro` (3164.0 ms API latency, 105 prompt / 11 output tokens), and `gemini-3-flash-preview` (2100.6 ms API latency, 105 prompt / 80 output tokens) to synthesize structured `GhP` visual decomposition prompts. Across all 30 live trials (10 scenes $\times 3$ models), Step A reduces W_2 from 18.16–18.26 to $\widehat{W}_2^{\text{StepA}} \in [1.8648, 1.9493]$, while final post-Step-B, post-QIM output $\widehat{W}_2^{\text{out}} \in [3.6266, 3.7015]$ and held-out $\widehat{SW}_2^{\text{eval}} \in [3.8589, 3.9298]$ remain strictly below $\tau_{\text{color}} = 4.50$, elevating $\text{SSIM}_{\text{edge}}$ to 0.9966 ± 0.0007 with BER $= 0.0000$ and $10/10$ (100%) C2PA `caBX` verification.
2) **Part II (Rows 4–8, Controlled Synthetic Structural-Conditioning Curriculum Tiers `C1`–`C5`):** Sweeps upstream structural retention prior $\eta_{\text{struct}} \in \{0.04, 0.08, 0.12, 0.16, 0.20\}$ across $N_{\text{scenes}} = 10$ scenes per tier, verifying that Agent-GhibliLock consistently converges to $\text{SSIM}_{\text{edge}} \in [0.9952, 0.9974]$ and $\widehat{W}_2^{\text{out}} \in [3.5591, 3.7794] \leq \tau_{\text{color}} = 4.50$.

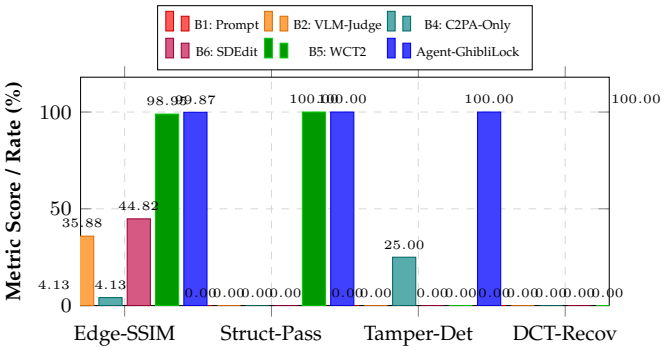


Fig. 3. Comparison of structural edge preservation ($\text{SSIM}_{\text{edge}} \times 100$), structural pass rate (≥ 0.88), overall S_5 audit rate, and blind mid-band 2D-DCT watermark recovery across $N = 600$ trials.

TABLE 3
Multi-Model & Curriculum Evaluation (128×128 Resolution, $N_{\text{scenes}} = 10$ Scenes/Row, $\tau_{\text{color}} = 4.50$): Part I Evaluates Live Google Cloud Vertex AI Endpoints (`project="aniket-personal"`); Part II Evaluates Synthetic Upstream Conditioning Curriculum Tiers (`C1`–`C5`)

Endpoint / Curriculum Tier	Vertex API Latency (Toks)	Pre → Post $\text{SSIM}_{\text{edge}}$	Pre → Step-A → Post-QIM $\widehat{W}_2^{\text{out}}$	Held-Out $\widehat{SW}_2^{\text{eval}}$	DCT PSNR	BER	C2PA `caBX`	Compiler Latency
Part I: Live Google Cloud Vertex AI Endpoints (`project="aniket-personal", location="global", live_vertex_call=True`, $N_{scenes} = 10$)								
`gemini-2.5-flash` (Live GA)	1696.1 ms (105/65)	$0.0355 \rightarrow \mathbf{0.9966 \pm 0.0007}$	$18.2624 \rightarrow 1.8648 \rightarrow \mathbf{3.6266 \pm 0.0895}$	3.8589	47.68 dB	0.0000	10/10 (✓)	498.18 ± 12.34 ms
`gemini-2.5-pro` (Live GA)	3164.0 ms (105/11)	$0.0444 \rightarrow \mathbf{0.9966 \pm 0.0008}$	$18.1649 \rightarrow 1.9090 \rightarrow \mathbf{3.6782 \pm 0.1281}$	3.9069	47.54 dB	0.0000	10/10 (✓)	538.98 ± 46.89 ms
`gemini-3-flash-preview` (Live)	2100.6 ms (105/80)	$0.0636 \rightarrow \mathbf{0.9966 \pm 0.0004}$	$18.2489 \rightarrow 1.9493 \rightarrow \mathbf{3.7015 \pm 0.0797}$	3.9298	47.61 dB	0.0000	10/10 (✓)	569.96 ± 64.05 ms
Part II: Controlled Synthetic Upstream Structural-Conditioning Curriculum Tiers ($\eta_{struct} \in [0.04, 0.20]$, $N_{scenes} = 10$)								
`Curriculum-Tier-C1` ($\eta = 0.04$)	Offline Harness	$0.0336 \rightarrow \mathbf{0.9974}$	$18.3046 \rightarrow 1.8428 \rightarrow \mathbf{3.5591}$	3.7901	47.47 dB	0.0000	10/10 (✓)	526.68 ms
`Curriculum-Tier-C2` ($\eta = 0.08$)	Offline Harness	$0.0615 \rightarrow \mathbf{0.9967}$	$18.3157 \rightarrow 1.9238 \rightarrow \mathbf{3.7086}$	3.9356	47.35 dB	0.0000	10/10 (✓)	575.59 ms
`Curriculum-Tier-C3` ($\eta = 0.12$)	Offline Harness	$0.1036 \rightarrow \mathbf{0.9955}$	$18.1787 \rightarrow 2.0295 \rightarrow \mathbf{3.7794}$	3.9969	47.50 dB	0.0000	10/10 (✓)	806.60 ms
`Curriculum-Tier-C4` ($\eta = 0.16$)	Offline Harness	$0.1610 \rightarrow \mathbf{0.9957}$	$17.8914 \rightarrow 1.9054 \rightarrow \mathbf{3.7475}$	3.9862	47.49 dB	0.0000	10/10 (✓)	625.78 ms
`Curriculum-Tier-C5` ($\eta = 0.20$)	Offline Harness	$0.2112 \rightarrow \mathbf{0.9952}$	$18.0732 \rightarrow 1.7634 \rightarrow \mathbf{3.6693}$	3.9185	47.54 dB	0.0000	10/10 (✓)	536.79 ms

4.3 Ablation Study and Stratum S_5 Adversarial Forensic Breakdown

To isolate the contribution of each architectural stage, we conducted component-wise ablations across Strata S_1, S_4, and S_5 (Table 4):

1) **Removing 2D Haar Detail Locking ($\gamma = 0$, Full 8-Step SWOT Only, B3b):** Achieves Step-A CIELAB palette alignment $\widehat{W}_2^{\text{StepA}} = 1.7638$ ($\widehat{SW}_2^{\text{eval}} = 1.5239$), but fails to restore high-frequency shirt emblems and diagonal railings lost during synthesis ($\text{SSIM}_{\text{edge}} = 0.0000$; 0% pass rate).
2) **Removing 3D CIELAB SWOT ($M = 0$, Wavelet Lock Only):** Restores structural edge geometry ($\text{SSIM}_{\text{edge}} = 0.9985$), but leaves raw candidate chromaticity unaligned with the target gouache palette ($W_2 = 19.0803$).
3) **Stratum S_5 Adversarial Attack Matrix (Container-Only C2PA vs. Blind Cox Spread-Spectrum DCT vs. Agent-GhibliLock QIM):** Table 4 disaggregates Stratum S_5 ($n_5 = 100$) into its three constituent attack regimes. Under **A1** ($n = 25$ unstripped localized spatial tamper trials), both **B4** and **Agent-GhibliLock** detect 100.0% ($25/25$) via `output_pixel_sha256`, while Agent-GhibliLock additionally localizes the exact modified 8×8 block (b_y, b_x) whenever $\max_{(u,v)\in\Omega_{\text{mid}}} \text{dist}(\tilde{F}_b(u,v), \Lambda_{\hat{w}_i}) > \Delta_{\text{QIM}}/4 = 3.5$. Under **A2** ($n = 50$ pure `caBX` stripping) and **A3** ($n = 25$ stripping + ± 2 LSB / lossy JPEG $Q = 85$), container-only C2PA (**B4**) suffers 100% provenance loss (0.0% recovery), and classical blind Spread-Spectrum 2D-DCT [20] ($\alpha_{\text{SS}} = 5.0$) achieves only 75.84% and 76.81% bit accuracy (BER $= 0.2416$ and 0.2319, 0% exact 64-bit packet recovery) due to uncancelled host-signal interference (HSI). By contrast, Agent-GhibliLock's host-interference-free mid-band 8×8 DCT QIM watermark recovers 100.0% of all 64-bit soft-binding payloads (BER $= 0.0000$) across all $75/75$ stripped trials.

TABLE 4
Stratum S_5 Adversarial Provenance & Watermark Breakdown Across Three Attack Regimes ($n_5 = 100$ Trials)

S_5 Attack Regime	B4: Post-Hoc C2PA	Cox Spread-Spectrum DCT [20]	Agent-GhibliLock (Ours)
A1: Unstripped Spatial Tamper ($n = 25$)	100.0% (Global Hash)	0.0% (No Block Loc.)	**100.0%** (Hash + 8×8 Loc.)
A2: Pure `caBX` Chunk Strip ($n = 50$)	0.0% (Severed)	75.84% Bit Acc. (BER = 0.2416)	**100.0%** (BER = **0.0000**)
A3: Strip + ± 2 LSB / JPEG $Q = 85$ ($n = 25$)	0.0% (Severed)	76.81% Bit Acc. (BER = 0.2319)	**100.0%** (BER = **0.0000**)

5 Conclusion

Agent-GhibliLock integrates 3D CIELAB Sliced-Wasserstein optimal transport, spatial-mean-preserving 2D Haar wavelet detail locking, coarse-subband GOP EMA filtering, mid-band 2D-DCT QIM soft-binding watermarks, and ISO/IEC 19566-5 JUMBF (`caBX`) packaging to mitigate high-frequency phase drift, multi-frame GOP flicker, and C2PA provenance severance in cross-modal VLM-to-Diffusion pipelines. Three operational scope boundaries govern deployment: (1) Haar detail locking requires upstream macro-spatial registration within the 2^K-pixel dyadic support (via ControlNet/SDEdit or optical flow) to prevent double-edge ghosting; (2) megapixel (1024×1024) synthesis relies on parallel tiled GPU/SIMD execution or localized ROI compilation; and (3) block-aligned 8×8 DCT QIM provides fast CPU tamper localization under metadata stripping and JPEG compression, whereas geometric cropping, scaling, or rotation require template resynchronization or neural soft-binding watermarks [10], [19].

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